An Internship Report

Of

Data Analytics- Job Simulation

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)

By

Roll Number of Student:- 2410030687

Name of Student: yugam

Batch: 2026-27

CANDIDATE'S DECLARATION

I, **Yugam** , hereby declare that this report presents my work completed as part of the **Deloitte Australia Data Analytics Job Simulation** through Forage.

This report is based on the tasks, data, tools, and outputs provided during the simulation. It describes the work performed, the methods used, and the learning outcomes gained during the completion of the simulation.

I have prepared this report to document my practical learning and experience in the areas of data analysis, data visualization, data classification, and business-oriented interpretation.

Student Name: Yugam Sachdeva
Roll No.: 2410030687

Signature: ___________________

ACKNOWLEDGEMENT

I would like to express my gratitude to **Deloitte Australia and Forage** for providing the Data Analytics Job Simulation opportunity.

The simulation provided practical exposure to data analysis and helped me understand how data can be used to solve business-related problems. Through the two tasks, I gained experience in using **Tableau for data visualization** and **Microsoft Excel for data classification and analysis**.

I am also thankful to the **School of Computer Science and Engineering, IILM University**, for providing the academic platform to document and present this learning experience.

INTRODUCTION & SIMULATION OVERVIEW

1. Introduction

Data analytics plays an important role in helping organizations understand information and make better business decisions. It involves examining data, identifying patterns, visualizing information, and using the results to answer business questions.

As part of the **Deloitte Australia Data Analytics Job Simulation**, I completed two practical tasks related to data analysis and forensic technology.

The simulation provided an opportunity to work with business-oriented datasets and understand how data can be transformed into useful insights.

2. Simulation Overview

The simulation consisted of two main tasks:

Task 1 – Data Analysis

The first task involved analysing **Daikibo's machine telemetry data** and creating a dashboard using **Tableau**.

Task 2 – Forensic Technology

The second task involved analysing **employee compensation data** and classifying equality scores using **Microsoft Excel**.

Tools Used

- **Tableau** – Data visualization and dashboard creation
- **Microsoft Excel** – Data classification and analysis

TASK 1: DATA ANALYSIS

Task 1 – Data Analysis

3.1 Background

Daikibo's technology team collected telemetry data from its four factories:

- **Daikibo Factory Meiyo** – Tokyo, Japan
- **Daikibo Factory Seiko** – Osaka, Japan
- **Daikibo Berlin** – Berlin, Germany
- **Daikibo Shenzhen** – Shenzhen, China

Each location had **nine types of machines** that sent telemetry messages every **10 minutes**.

The data was collected for one month, **May 2021**, and was provided as a single JSON file.

3.2 Objective

The main objective of this task was to analyse the telemetry data and use visualization to answer two questions:

1. **In which location did machines break the most?**
2. **What are the machines that broke most often in that location?**

The analysis was performed using **Tableau**.

TASK 1: TABLEAU ANALYSIS

3.3 Analysis Performed

I used Tableau to explore and visualize the Daikibo telemetry data.

The analysis focused on comparing **unhealthy machine readings** across different factory locations and device types.

A dashboard was created containing visualizations for:

- Factory
- Device Type
- Unhealthy readings

The factory-level visualization helped compare the number of unhealthy readings across the four Daikibo locations.

3.4 Dashboard

[INSERT YOUR TABLEAU DASHBOARD SCREENSHOT HERE]

Figure 1: Tableau Dashboard for Daikibo Telemetry Data Analysis

3.5 Main Finding

From the dashboard analysis, **Daikibo Factory Seiko** showed the highest number of unhealthy readings among the four factory locations.

The device-type visualization can then be used to explore which machine types contributed most to the unhealthy readings in the selected location.

3.6 Learning from Task 1

Through this task, I learned:

- How to analyse telemetry data.
- How to compare data across multiple locations.
- How to create charts using Tableau.
- How to combine visualizations into a dashboard.
- How data visualization can make large datasets easier to understand.
- How dashboards can help identify areas that require attention.

4. Task 2 – Forensic Technology

4.1 Background

The second task focused on helping a team investigate **unfair pay** using employee compensation data.

The provided Excel file contained information related to employee compensation and equality.

The dataset included three main columns:

1. **Factory**
2. **Job Role**
3. **Equality Score**

4.2 Equality Score

The **Equality Score** was an integer ranging from **−100 to +100**, where **0 represents the ideal score**.

The purpose of the task was to make the equality information easier to interpret by creating an additional classification column.

4.3 Objective

The objective was to add a fourth column called:

Equality Class

The equality score was classified into three categories:

- **Fair**
- **Unfair**
- **Highly Discriminative**

The work was completed using **Microsoft Excel**.

TASK 2: EXCEL CLASSIFICATION

4.4 Equality Classification

The task instructions provided the following classification:

Equality Class	Classification
Fair	Within ±10

Unfair	Beyond ±10
Highly Discriminative	Beyond ±20

Examples provided in the task:

- **10 → Fair**
- **−9 → Unfair**
- **−30 → Highly Discriminative**

Note: The supplied task instructions contain an apparent overlap/inconsistency because −9 is shown as Unfair even though it falls within ±10. The examples and categories above are retained as provided in the task material.

4.5 Work Performed

I added the required **Equality Class** column to the Excel dataset.

This classification made the equality scores easier to interpret and helped group the data according to the level of pay equality.

Completed Excel Data

[INSERT SCREENSHOT OF YOUR COMPLETED EXCEL SHEET HERE]

Figure 2: Completed Equality Table with Equality Class

4.6 Outcome

The classification of equality scores helped identify potential pay equality issues by grouping the records into meaningful categories.

This task provided practical experience in using Excel to organize and classify business data.

SKILLS & LEARNING OUTCOMES

5. Skills Developed

The simulation helped me develop practical skills in the following areas:

Data Analysis

I learned how to examine structured datasets and use the available information to answer specific business questions.

Data Visualization

Using Tableau, I learned how to represent data visually and create dashboards for easier interpretation.

Microsoft Excel

I gained experience in organizing data and creating an additional classification column based on equality scores.

Data Classification

I learned how numerical values can be grouped into meaningful categories to support analysis.

Business Interpretation

The tasks helped me understand how data analysis can be used to identify patterns and support business-related conclusions.

5.1 Overall Learning

The simulation helped me understand that data analytics is not only about working with numbers. It also involves presenting information clearly, identifying useful patterns, and converting data into conclusions that can support decision-making.

The overall workflow I experienced was:

Data → Analysis → Visualization/Classification → Insight

CONCLUSION + REFERENCES

6. Conclusion

The **Deloitte Australia Data Analytics Job Simulation** provided practical exposure to two important areas of data analytics.

In **Task 1**, I analysed Daikibo's machine telemetry data and created a Tableau dashboard to compare unhealthy machine readings across different factory locations and device types.

In **Task 2**, I worked with employee compensation data and used Excel to classify equality scores into different categories as part of an unfair-pay investigation.

Overall, the simulation improved my understanding of **data analysis, data visualization, Excel-based classification, and business-oriented interpretation**.

It also helped me understand how data can be transformed into meaningful information that can support business decisions.

7. References

1. **Deloitte Australia Data Analytics Job Simulation – Forage.**
2. **Daikibo Telemetry Data Analysis task material provided in the simulation.**
3. **Equality Table / Forensic Technology task material provided in the simulation.**
4. **Tableau – Data Visualization and Dashboard Creation.**
5. **Microsoft Excel – Data Analysis and Classification.**

Deloitte.

yugam sachdeva

Data Analytics Job Simulation

Certificate of Completion

June 22nd, 2026

Over the period of June 2026, yugam sachdeva has completed practical tasks in:

Data analysis
Forensic technology

Tina McCreery
Chief Human
Resources Officer,
Deloitte

Enrolment Verification Code P4oedBBHhgbTNP9ox | User Verification Code 6a388497e67d5b12bbba061e | Issued by Forage